

Part-A Fundamentals

Q1. What is Data Analysis?

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to discover useful information, identify patterns, and support decision-making. It helps organizations make informed business decisions using data-driven insights.

Q2. Steps of Data Science Project:

- Business Understanding
- Data Understanding
- Data Preparation (EDA)
- Model
- Evaluation
- Deployment

Q3. ML Problem Statement

The objective is to predict whether a customer will churn (leave the service) based on customer demographics, account information, and purchase behavior. This is a supervised binary classification problem where the target variable is Churn (Yes/No).

Q4. What are Tensors?

In the context of machine learning, think of a Tensor as a multi-dimensional array. It is the fundamental data structure used by modern AI libraries like TensorFlow or PyTorch.

- 0-D Tensor: A scalar (a single number, e.g., 5).

- 1-D Tensor: A vector (a list of numbers, e.g., ``).
- 2-D Tensor: A matrix (a table of numbers).
- 3-D+ Tensor: A cube or higher-dimensional volume of numbers.

Example in NumPy:

```
import numpy as np
```

```
# Scalar
```

```
scalar = np.array(10)
```

```
# Vector
```

```
vector = np.array([1,2,3])
```

```
# Matrix
```

```
matrix = np.array([  
    [1,2],  
    [3,4]  
])
```

```
# 3D Tensor
```

```
tensor = np.array([  
    [[1,2],[3,4]],  
    [[5,6],[7,8]]  
])
```

```
print(tensor.shape)
```

```
# (2,2,2)
```